UNIT-2

UNIT-II

CHAPTER 1

Introductory Remarks

Language has a very important social purpose, because it is mainly used for linguistic communication. (Communication is quite possible without the use of language. For example, a dog barks and informs its master of the approach of a stranger. A child cries and informs its mother that it is hungry, thirsty or uncomfortable. In both the examples we have cited above, communication *does* take place, but no language is used. In this book we will deal with **linguistic** communication.)

A language can be used in two ways for the purposes of communication. It can be *spoken* or *written*. In other words, we can communicate, using the same language, using the spoken *medium* or the written *medium*. (i) (ii)

The medium of speech is more important than the medium of writing. This is because speech comes first in the history of any language community—in fact it came centuries before writing in the history of any language community. Secondly, speech comes first in the history of any individual. We started speaking long before we started writing. Thirdly, speech as a medium of communication is used much more than the medium of writing. Fourthly, written language is only an attempt to represent, using marks on paper, the sounds used in spoken language. In every language, a letter of the alphabet represents a particular sound. (In a language like English, for example, a letter of the alphabet at times represents more than one sound. See Chapter 6.) Lastly, modern technology has contributed tremendously to the importance of speech—modern inventions like the telephone, the radio, the tape recorder and several such devices have raised problems of communication primarily concerned with speech. (iii) (iv) (v) (vi) (vii)

Linguistics is a systematic study of language. Phonetics is a branch of linguistics and it is the branch dealing with the medium of speech. It deals with the production, transmission and reception of the sounds of human speech. In the following chapters we shall look into some of these. (viii)

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CHAPTER 2

The Air-Stream Mechanisms

For the production of speech sounds we need an air-stream. This will become evident if we do a simple experiment. Say a prolonged *ah* in a whisper. If you put your palm in front of your mouth while saying *ah* in whisper, your palm will feel the warm air getting out of your mouth. It becomes evident, therefore, that while you are saying *ah*, air gets out of your mouth. Now say *baa* several times, again in a whisper and then say *ah* and *baa* alternatively several times. You will see that the air that escapes from your mouth is blocked temporarily when you close your lips for producing the **b** of *baa* and when you open your lips the air continues to flow out of your mouth. In other words, the air that flows out of your mouth is *modified* into speech sounds by the action of certain organs of your body (These are commonly referred to as **organs of speech** and these are discussed elaborately in the next chapter). Thus an air-stream is the basis of speech sounds. An air-stream can also be called a moving current of air. For the articulation of most speech sounds of most languages, **lung-air** is used. The air that we breathe out is modified into speech sounds. Let us, therefore, look at the respiratory mechanism in some detail.

The respiratory system consists of the lungs, the muscles of the chest and the windpipe (also called the **trachea**).

The lungs are spongy bodies. They are made up of small sacs called the **alveoli.** It is in these sacs that the blood is cleaned of its carbon-dioxide and provided with fresh oxygen from the outer-air. Air is supplied to the alveoli by small tubes called the **bronchioles.** The bronchioles come together into two large tubes called the **bronchi,** one situated on the right and the other on the left. The bronchi join the **trachea** or the **windpipe.** It is through the trachea that the air that we breathe in passes through the throat into the lungs. The act that is commonly known as respiration involves two processes—taking outer air into the lungs (called **inspiration**) and throwing out air from the lungs into the outer atmosphere (called **expiration**).

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It is the expiratory lung-air, that is, the air that we breathe out, that is the basis for the articulation of most speech sounds. For this reason the air-stream involving lung-air is called **pulmonic air-stream.**

An air-stream is produced by an **air-stream mechanism.** An air-stream mechanism works in exactly the same way as a flit-gun. In a flit-gun, a body of liquid (or air if it is empty) is moved by working an apparatus (which is part of the flit-gun) called a plunger. When the plunger is moved in one direction, the liquid (or air) is pulled in and when it is moved in the opposite direction, the liquid (or air) is pushed out. When we speak, something like the plunger in a flit-gun is in operation, working in the same way as the plunger does in a flit-gun. This is called the *initiator*. The air-stream is set in motion by the initiator, just as the liquid in a flit-gun is set in a motion by the plunger.

There are three main air-stream mechanisms. These are called **pulmonic, glottalic** and **velaric** air-stream mechanisms. Since a very large percentage of speech sounds in all the languages that are spoken in the world use a pulmonic air-stream mechanism, let us consider it first.

The pulmonic air-stream mechanism consists of the lungs and the respiratory muscles. The **walls of the lungs** act as the initiator. They are moved by the respiratory muscles so that air is drawn into the lungs or pushed out of them. When the air-stream mechanism is used to push air out, it is called **egressive** and when it is used to draw air in, it is called **ingressive.**

As we said earlier, most speech sounds make use of a pulmonic egressive air-stream mechanism. In fact **all** the sounds of English and most Indian languages (Sindhi is an exception; we shall discuss this a little later in this chapter) are produced with this air-stream mechanism. It is possible to speak using a pulmonic **ingressive** air-stream mechanism, but no language uses this air-stream mechanism for the production of speech sounds. We use this air-stream machanism for yawning and snoring, but not for speaking.

Two other air-stream mechanisms are possible. One is called **glottalic air-stream mechanism.** The closed glottis (see Chapter 3) acts as the initiator for this and the air in the pharynx (see Figure 1 in Chapter 3) is used. For this reason some linguists refer to this air-stream mechanism as **pharyngeal.** Both egressive and ingressive glottalic air-stream mechanisms are used by some languages of the world for the articulation of speech sounds. Among Indian languages,

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Sindhi has sounds articulated with a glottalic ingressive air-stream mechanism. (These sounds are called **implosives.**)

The third air-stream mechanism is commonly referred to as **velaric air-stream mechanism.** The back of the tongue (see Chapter 3) is the initiator and the air in the mouth is set in motion during this air-stream mechanism. For this reason, this is referred to as the **oral** air-stream mechanism. Sounds produced with a velaric **ingressive** mechanism exist in several African languages; sounds produced with a velaric egressive air-stream mechanism do not seem to exist in any language.

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CHAPTER 3

The Organs of Speech

We saw in the last chapter that for the production of speech sounds we need an air-stream mechanism and that most sounds of most languages in the world are produced with a *pulmonic egressive air-stream mechanism.* The air that we breathe out comes out of the lungs. Before it gets out into the outer atmosphere, various organs

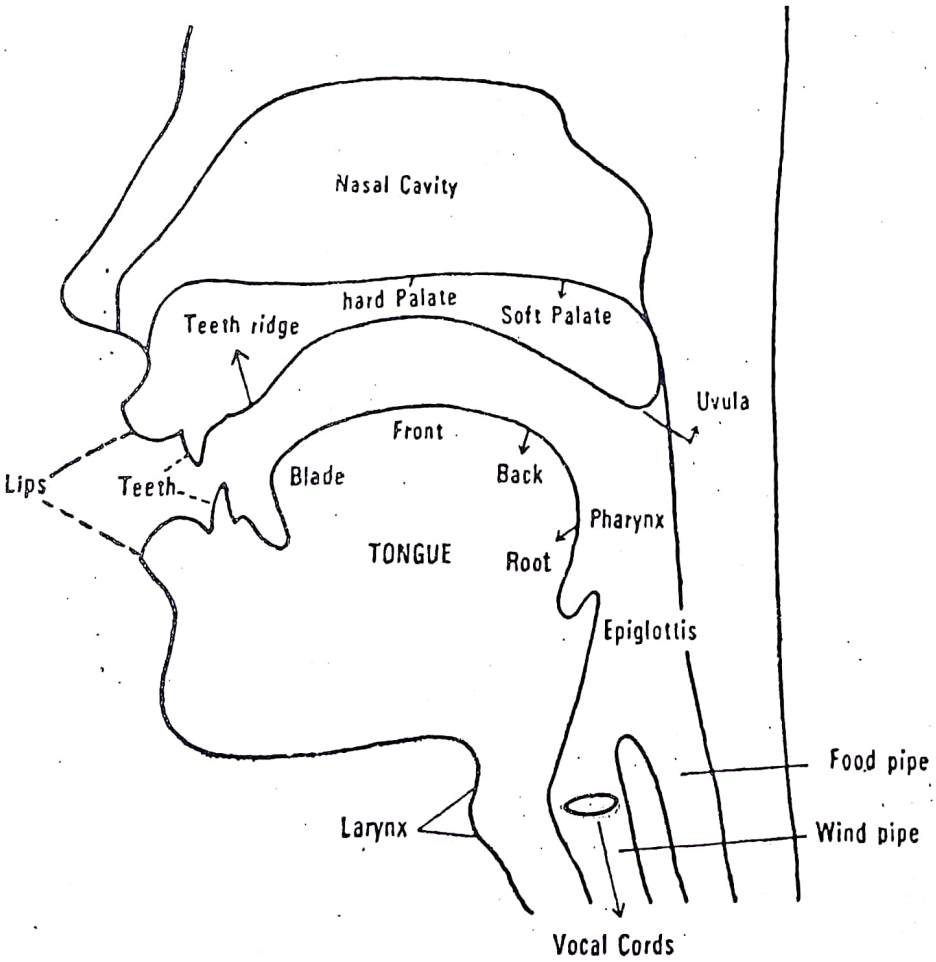

Fig. 1

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in our body convert it into speech sounds. These organs are called the *organs of speech*.

The expression *organs of speech* is not a happy one because all the organs that are described thus have functions other than the production of speech. The lungs, the vocal cords (these are described a little later in this chapter), the tongue, the teeth and the lips are some of the organs that are described commonly in phonetic literature as organs of speech. The lungs are necessary for breathing, the vocal cords shut tightly and thus prevent the food from entering the windpipe, the tongue is used for differentiating tastes, the teeth are used for chewing food, and so on. In addition to these primary functions, these organs are also involved in the production of speech and hence these are referred to as organs of speech. For want of a better expression, we shall continue to refer to these as organs of speech in this book.

Figure 1 is a vertical cross-section of the human head and neck. The organs of speech can be divided into the following three groups:

1. *The respiratory system:* This comprises the lungs, the muscles of the chest and the windpipe (the windpipe is also called the *trachea*).
2. *The phonatory system:* This comprises the larynx (see Figure 1).
3. *The articulatory system:* This comprises the nose, the teeth, the tongue, the roof of the mouth and the lips (see Figure 1).

The Respiratory System

The action of the lungs and the muscles of the chest has been described in the previous chapter.

The Phonatory System

The Larynx: The larnyx is what is commonly called the *Adam's apple*, situated at the top of the windpipe. The air from the lungs has to come out through the windpipe and the larynx. In the larynx are situated a pair of lip-like structures. These are called the *vocal cords* and these are placed horizontally from front to back. They are attached in front and can be separated at the back. The opening between the cords is called the *glottis* (see Figure 2).

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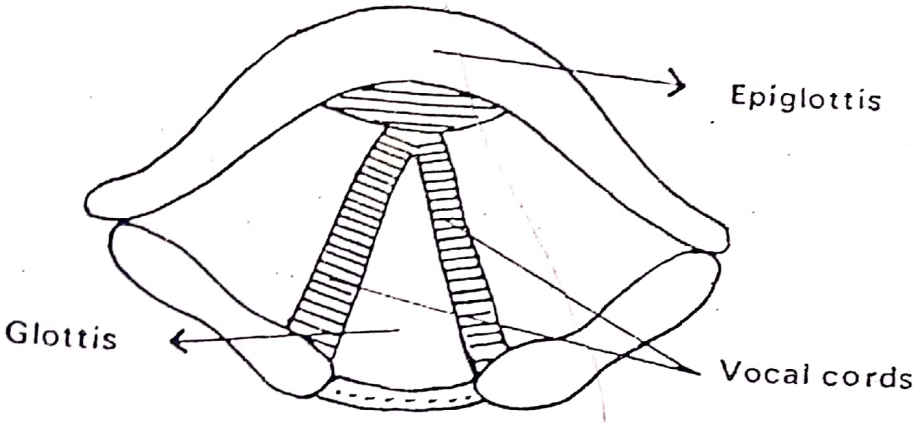

Fig. 2

The vocal cords can be opened and closed (because they can be separated at the back) and when the two cords come very close to each other, the glottis will be shut completely. In fact when we swallow food or water, the vocal cords shut the glottis and thus prevent the food or water from entering the windpipe.

When we breathe in and out, the vocal cords are *drawn wide apart* and thus the glottis is open. The air enters the lungs or gets out of the lungs through the wide open glottis. When we produce some speech sounds, the vocal cords are wide apart and the glottis is open. Such sounds produced with a wide-open glottis are called *voiceless sounds* or *breathed sounds* (the latter term is used because this is the position of the glottis for breathing). The first sounds in the English words *peel, ten, keen , chin, fine, thin, seen, shine* and *hat* are *voiceless* sounds.

During the production of certain speech sounds, the vocal cords are *loosely held together* and the pressure of the air from the lungs makes them open and close rapidly. This is called the *vibration* of the vocal cords and the sounds produced when the vocal cords vibrate are called *voiced* sounds. All the sounds in the English words *bead, deed, girl, judge, vine, then, zoo, measure, need, wing, red, yard* and *well* are voiced sounds.

Figures 3 and 4 given below represent the positions of the vocal cords during the production of voiceless and voiced sounds respectively.

The vibration of the vocal cords is important for another factor, too. The rate at which the vocal cords vibrate is called the *frequency*

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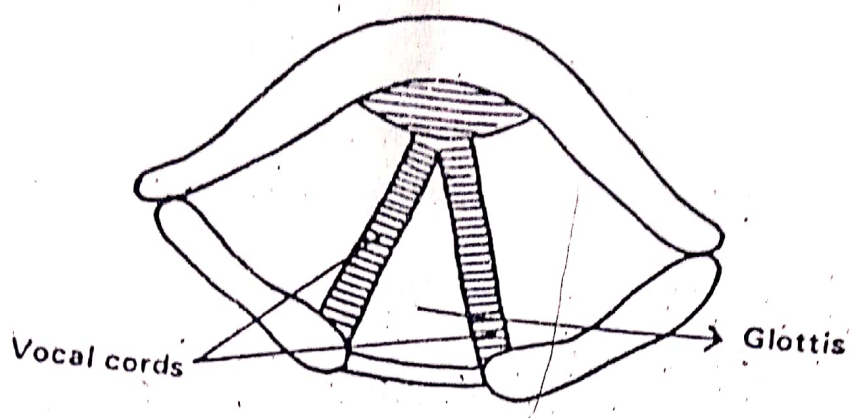

Fig. 3
Vocal cords wide apart and the glottis fully open—position for breath and during the production of voiceless sounds.

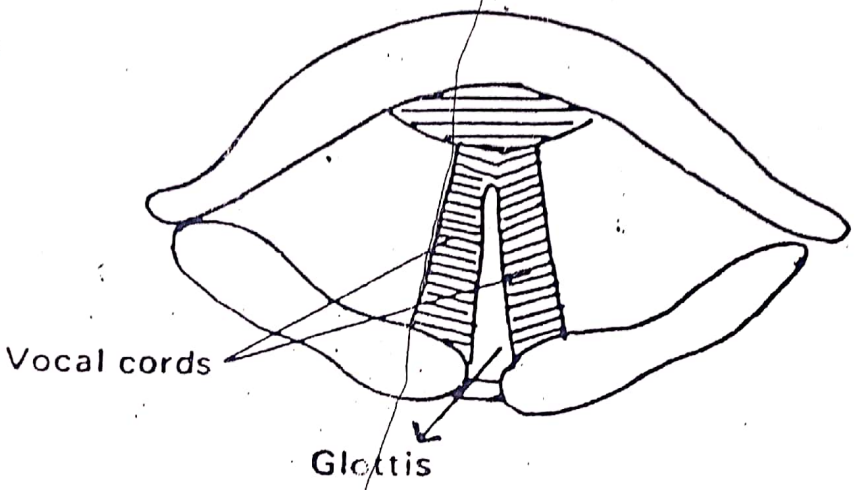

Fig. 4
Vocal cords kept loosely together—position for vibration during the production of voiced sounds.

of vibration of the vocal cords and this determines the *pitch* of the voice. We shall discuss pitch in a later chapter.

We can decide whether a sound is voiceless or voiced (i.e., whether the vocal cords vibrate or not during its production) by conducting one of two simple experiments. If we produce a sound and place our fingers lightly on the Adam's apple during its production, our fingers will 'feel' the vibration of the vocal cords if the sound produced is a *voiced sound*. If the sound produced is voiceless, our fingers will 'feel' nothing. To take an example, let us produce a prolonged **ssss** (i.e., a prolonged hissing sound) with our fingers on the Adam's apple. Then if we produce a prolonged **zzzz** (i.e., a prolonged buzzing sound), during the production of **zzzz** our fingers will 'feel' the vib-

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ration of the vocal cords. They will 'feel' no such thing during the production of ssss. The *hissing* sound is therefore *voiceless* and the *buzzing* sound *voiced*.

There is another experiment, which some people think is easier than the one described above. If we produce a voiced sound with our palms placed gently over our ears, a buzz will be heard, whereas no such buzz will be heard if the sound produced is voiceless. Try the experiment with a prolonged **ssss** and then with a prolonged **zzzz**. Make sure that you hear a buzz when you say **zzzz** and no such buzz when you say ssss. Then try the experiment with other sounds like **ffff** and **vvvv**. You should hear a buzz when you say **vvvv** and no such buzz when you say **ffff**.

The Articulatory System

Let us now look at the area called the roof of the mouth (see Figure 1). This comprises the *teeth-ridge*, the *hard palate*, the *soft palate* and the *uvula*.

The convex bony part of the roof of the mouth which lies immediately behind the upper front teeth is called the *teeth-ridge*. It is also called the *alveolar ridge* or the *alveolum*. Immediately after the teeth ridge, the roof of the mouth becomes concave and it is hard and bony. This bony concave surface is called the *hard palate*. The roof of the mouth then suddenly becomes soft and fleshy. This is called the *soft-palate* or the *velum*. The fleshy structure hanging loose at the extreme end of the roof of the mouth is called the *uvula*.

Of the organs comprising the roof of the mouth, the soft palate or the velum has to be considered in some detail. The soft palate acts like a valve in opening and closing the *nasal passage of air*. If the soft palate is *raised* so that it touches the back wall of the pharynx, the passage into the nose is closed. The air then cannot escape through the nose at all (see Figure 5). Sounds during the production of which the air escapes only through the mouth are called *oral* sounds. All the sounds in the English words *peel*, *bag*, *saw*, *date* and *bush* are *oral* sounds. The closure of the nasal passage of air by raising the velum is called *velic closure*.

If the soft palate is lowered, the passage into the nose is opened. With the soft palate lowered, if the passage into the mouth is blocked (either by closing the lips or by making the tongue touch firmly some part of the roof of the mouth) the air from the lungs will escape into

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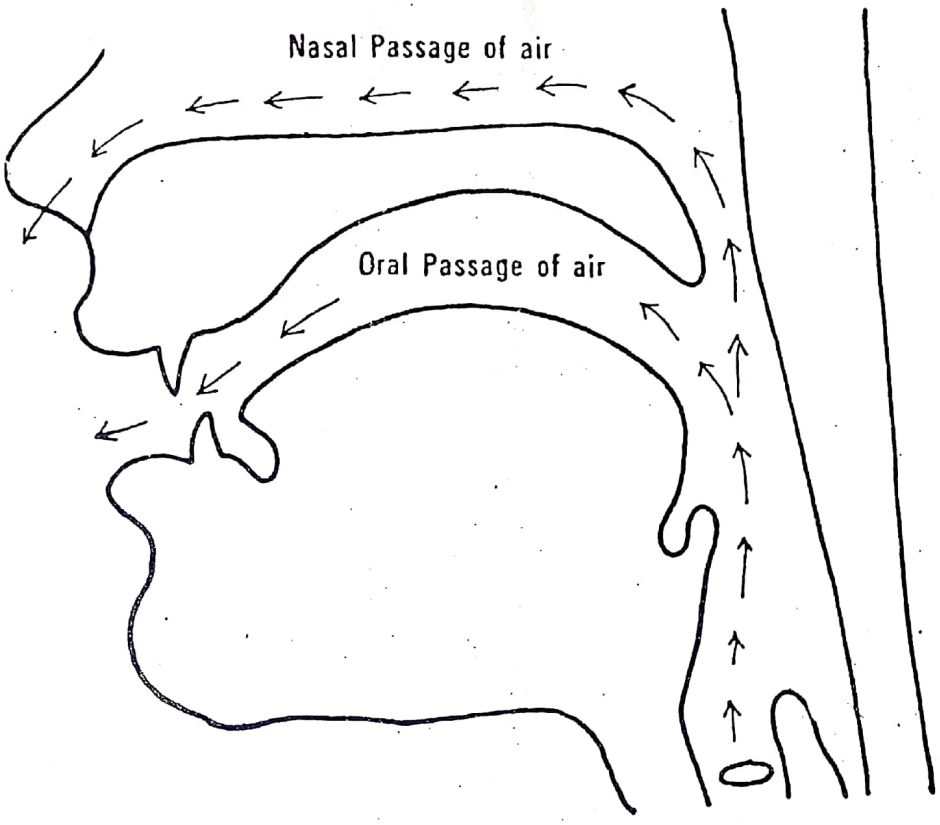

Fig. 7
Production of nasalised sounds. The soft palate is lowered, thus opening the nasal passage of air. There is no blockage of the oral passage of air. The air escapes simultaneously through the nose and the mouth (indicated by arrows). There is neither velic closure nor any oral closure.

which lies opposite the hard palate when the speech organs are at rest. Beyond the front is the *back* of the tongue, which lies opposite the soft-palate when the speech organs are at rest (see Figure 1). Even beyond the back of the tongue is its *root*.

The lips: The lips play their part in the articulation of certain consonants. For example, the first sounds in the English words *pill*, *bill* and *mill* are produced with the lips tightly shut. Also, the lips play an important part during the articulation of vowel sounds. (This will be discussed in detail in Chapter 5.)

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EXERCISES

I. Say whether the following statements are true or false:

1. The lungs and the muscles of the chest play an important part in the production of speech sounds.
2. The air escapes from the lungs into the outer atmosphere through the glottis.
3. The glottis is shut completely during normal breathing.
4. The vocal cords are situated in the larynx.
5. The vocal cords are joined together at both ends.
6. The space between the vocal cords is called the glottis.
7. The vocal cords are kept loosely together during the production of voiceless sounds.
8. The glottis is wide open during the production of voiced sounds.
9. The state of the glottis is the same during normal breathing and during the articulation of voiceless sounds.
10. The frequency of vibration of the vocal cords determines the pitch of our voice.
11. The glottis is completely shut while swallowing food or drink.
12. The soft palate is not capable of independent movement.
13. The soft palate plays an important part in closing and opening the oral passage of air.
14. The soft palate is also called the velum.
15. The soft palate touches the back wall of the pharynx when we articulate oral sounds.
16. All sounds in all languages have to be either purely oral or purely nasal.
17. If the soft palate is lowered, the air cannot escape through the mouth.
18. If the soft palate is raised, the air cannot escape through the nose.
19. If the soft palate is lowered, the oral passage of air is shut off completely.
20. The front of the tongue lies opposite the hard palate when the speech organs are at rest.
21. The bony convex part of the roof of the mouth that lies immediately after the upper front teeth is called the teeth ridge.
22. The teeth ridge is also called the alveolum.

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why spoken medium is important than the written med.

The medium of speech is more important than the medium of writing because

→ Speech comes first in the history of any language community.

→ speech comes first in the history of any individual

→ speech is a medium of communication which is used more than the ~~written~~ written medium.

→ Language in written mode is just a representation of sounds used in spoken language.

what is a Phonetics ?

Linguistics is a systematic study of language and Phonetics is the branch of linguistics. It deals with the medium of speech. It deals with the production, transmission and reception of the sounds human speech sounds.

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The Respiratory System :-

The respiratory system consist of the lungs, the muscles of the chest and the windpipe also called as trachea. The respiration involves two process taking outer air into the lungs and throwing out air from the lungs into the outer atmosphere. The air that we breathe out is responsible for the articulation of most speech sounds. The air-stream involving lung-air is called as pulmonic air-stream.

There are three main air-stream mechanisms. They are Pulmonic, glottalic and velaric air-stream mechanisms. A large percentage of speech sounds in all the languages that are spoken in the world use a pulmonic air-stream mechanism.

The Pulmonic air-stream mechanism consist of the lungs and the respiratory muscles. The walls of the lungs act as the initiator. They are moved by the respiratory muscles so that air is drawn into the lungs or pushed out of it. When the air-stream mechanism is used to push air out, it is called egressive and when it is used to draw air in, it is called ingressive.

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Difference between Vowel and Consonant Sounds

Speech sounds are very broadly divided into two categories, namely, Vowels and consonants. If we say English word shoe, we realize that this word is made up of two sounds, one represented by the letters 'sh' and the other represented by the letter 'oe'. When we produce the sound represented by the letters 'sh' slowly, we see that during the production of this sound, the air escapes through the mouth with friction. On the other hand, when we produce the sound represented by the letter 'oe' the air escapes through the mouth freely and we do not hear any friction. The sound that is represented by the letters 'sh' in the 'shoe' is a consonant sound and the sound represented by 'oe' in the word 'shoe' is a vowel sound. All sounds during the production of which we hear friction are consonants, but not all consonants are produced with friction.

In words like she, shoe, show & ship, see that the sounds represented by the letters e, oe, y, ow, i and ou are vowel sounds because air escapes freely without any friction. Similarly in words like shoe, see, zoo & who we hear friction during the production of sound represented by the letter sh, s, z and wh are consonant. But each sound differs from each other.

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Description of consonants [Place of Articulation]

A consonant is usually described, stating whether it is voiceless or voiced, its place of articulation and its manner of articulation. [Manner of articulation refers to the structure involved] and such Plosive, affricate are labels given to consonants according to their manner of articulation. [Place of articulation just means the two articulators involved in the production of a consonant. Consonants can be described according to their place of articulation. The Places of articulation that we frequently come across are the following.

a) Bilabial :- The two lips are the articulators. The initial sounds in the English words Pile, bile & mile are bilabial sounds.

b) Labio-dental :- The active articulator is the lower lip and the passive articulators are the upper front teeth. The initial sounds in english words as 'fine' and 'vine'

c) Dental :- The tip of the tongue is the active articulator and the upper front teeth are the Passive articulators. The initial sounds in the English 'thin' and 'then'. /θ/ /ð/

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d) Alveolar :- The tip or the blade of the tongue is the active articulator and the teeth ridge is the passive articulator. The initial sound in the English words tin, din, near, sin, zip & love.

e) Past-Alveolar :- The tip of the tongue is the active articulator and the past of the roof of the mouth that lies immediately behind the teeth ridge is the passive articulator. The sound represented by the letter r as in 'try' and 'dry'.

f) Palato-Alveolar :- The tip of the tongue and blade of the tongue are the active articulators and the teeth ridge is the passive articulator. Simultaneously, the front of the tongue is raised in the direction of the hard palate. The initial sounds in the English words chup, cheap, 'jump' and 'jeep'.

Palatal :- The front of the tongue is the active articu and the hard palate is the passive articulator. The initial sound in the English word 'Yes' & 'Yog

Velar :- The back of the tongue is the active articulator and the soft palate is the passive articulator. The fi sound in English words back bag, bang & king.

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Glottal :→ Glottal sounds are produced at the glottis and the two vocal cords are the articulators. The initial sound in the English word hat is an example of a glottal sound.

Place → / Manner ↓	Bilabial	Labio-Dental	Dental	Alveolar	Post Alveolar	Palato Alveolar	Palatal	Velar	Glo.
Nasal	m			n				ŋ	
Plosive	p b			t d				k g	
Africate						tʃ dʒ			
Fricative		f v	θ ð	s z		ʃ ʒ			h
Lateral				l					
Approximant	w				ɹ		j		

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2 The production of speech sounds

2.1 Articulators above the larynx

All the sounds we make when we speak are the result of muscles contracting. The muscles in the chest that we use for breathing produce the flow of air that is needed for almost all speech sounds; muscles in the **larynx** produce many different modifications in the flow of air from the chest to the mouth. After passing through the larynx, the air goes through what we call the **vocal tract**, which ends at the mouth and nostrils; we call the part comprising the mouth the **oral cavity** and the part that leads to the nostrils the **nasal cavity**. Here the air from the lungs escapes into the atmosphere. We have a large and complex set of muscles that can produce changes in the shape of the vocal tract, and in order to learn how the sounds of speech are produced it is necessary to become familiar with the different parts of the vocal tract. These different parts are called **articulators**, and the study of them is called **articulatory phonetics**.

Fig. 1 is a diagram that is used frequently in the study of phonetics. It represents the human head, seen from the side, displayed as though it had been cut in half. You will need to look at it carefully as the articulators are described, and you will find it useful to have a mirror and a good light placed so that you can look at the inside of your mouth.

i) The **pharynx** is a tube which begins just above the larynx. It is about 7 cm long in women and about 8 cm in men, and at its top end it is divided into two, one

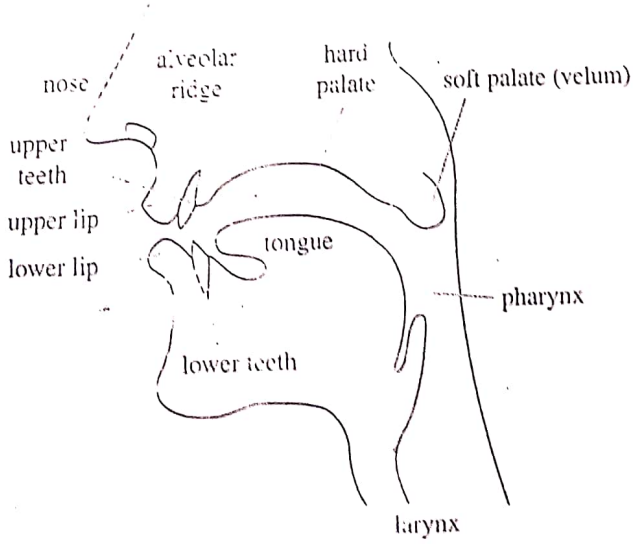

Fig. 1 The articulators

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part being the back of the oral cavity and the other being the beginning of the way through the nasal cavity. If you look in your mirror with your mouth open, you can see the back of the pharynx.

ii) The **soft palate** or **velum** is seen in the diagram in a position that allows air to pass through the nose and through the mouth. Yours is probably in that position now, but often in speech it is raised so that air cannot escape through the nose. The other important thing about the soft palate is that it is one of the articulators that can be touched by the tongue. When we make the sounds k, g the tongue is in contact with the lower side of the soft palate, and we call these **velar** consonants.

iii) The **hard palate** is often called the "roof of the mouth". You can feel its smooth curved surface with your tongue. A consonant made with the tongue close to the hard palate is called **palatal**. The sound j in 'yes' is palatal.

iv) The **alveolar ridge** is between the top front teeth and the hard palate. You can feel its shape with your tongue. Its surface is really much rougher than it feels, and is covered with little ridges. You can only see these if you have a mirror small enough to go inside your mouth, such as those used by dentists. Sounds made with the tongue touching here (such as t, d, n) are called **alveolar**.

v) The **tongue** is a very important articulator and it can be moved into many different places and different shapes. It is usual to divide the tongue into different parts, though there are no clear dividing lines within its structure. Fig. 2 shows the tongue on a larger scale with these parts shown: **tip, blade, front, back** and **root**. (This use of the word "front" often seems rather strange at first.)

vi) The teeth (upper and lower) are usually shown in diagrams like Fig. 1 only at the front of the mouth, immediately behind the lips. This is for the sake of a simple diagram, and you should remember that most speakers have teeth to the sides of their mouths, back almost to the soft palate. The tongue is in contact with the upper side teeth for most speech sounds. Sounds made with the tongue touching the front teeth, such as English θ, ð, are called **dental**.

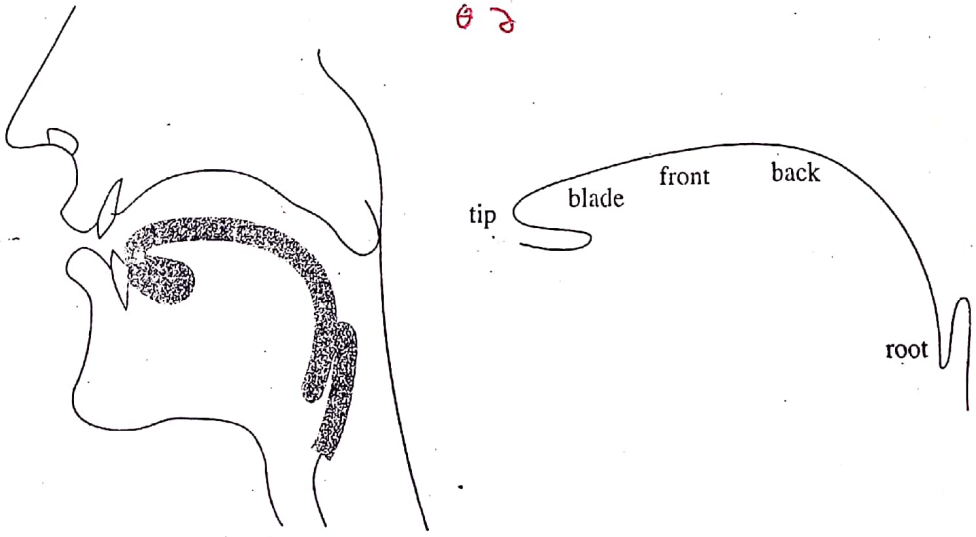

Fig. 2 Subdivisions of the tongue

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vii) The lips are important in speech. They can be pressed together (when we produce the sounds p, b), brought into contact with the teeth (as in f, v), or rounded to produce the lip-shape for vowels like u:. Sounds in which the lips are in contact with each other are called **bilabial**, while those with lip-to-teeth contact are called **labiodental**.

The seven articulators described above are the main ones used in speech, but the: are a few other things to remember. Firstly, the larynx (which will be studied in Chapter could also be described as an articulator – a very complex and independent one. Secondl the jaws are sometimes called articulators; certainly we move the lower jaw a lot in spea: ing. But the jaws are not articulators in the same way as the others, because they cannc themselves make contact with other articulators. Finally, although there is practically noth ing active that we can do with the **nose** and the nasal cavity when speaking, they are a ve: important part of our equipment for making sounds (which is sometimes called our voca **apparatus**), particularly nasal consonants such as m, n. Again, we cannot really describ the nose and the nasal cavity as articulators in the same sense as (i) to (vii) above.

2.2 Vowel and consonant

Difference between vowels & consonants.

The words **vowel** and **consonant** are very familiar ones, but when we study th sounds of speech scientifically we find that it is not easy to define exactly what they mea: The most common view is that vowels are sounds in which there is no obstruction to the flow of air as it passes from the larynx to the lips. A doctor who wants to look at the bac of a patient's mouth often asks them to say "ah"; making this vowel sound is the best wa of presenting an unobstructed view. But if we make a sound like s, d it can be clearly felt that we are making it difficult or impossible for the air to pass through the mouth. Mo: people would have no doubt that sounds like s, d should be called consonants. Howeve there are many cases where the decision is not so easy to make. One problem is that som English sounds that we think of as consonants, such as the sounds at the beginning of th words 'hay' and 'way', do not really obstruct the flow of air more than some vowels d Another problem is that different languages have different ways of dividing their sounc into vowels and consonants; for example, the usual sound produced at the beginning c the word 'red' is felt to be a consonant by most English speakers, but in some other lan guages (e.g. Mandarin Chinese) the same sound is treated as one of the vowels.

(i)

(ii) If we say that the difference between vowels and consonants is a difference in the wa that they are produced, there will inevitably be some cases of uncertainty or disagreemen this is a problem that cannot be avoided. It is possible to establish two distinct groups c sounds (vowels and consonants) in another way. Consider English words beginning wit the sound h; what sounds can come next after this h? We find that most of the sounc we normally think of as vowels can follow (e.g. e in the word 'hen'), but practically non of the sounds we class as consonants, with the possible exception of j in a word such a 'huge' hjuːdʒ. Now think of English words beginning with the two sounds bɪ; we fin many cases where a consonant can follow (e.g. d in the word 'bid', or l in the word 'bill'

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but practically no cases where a vowel may follow. What we are doing here is looking at the different contexts and positions in which particular sounds can occur; this is the study of the **distribution** of the sounds, and is of great importance in phonology. Study of the sounds found at the beginning and end of English words has shown that two groups of sounds with quite different patterns of distribution can be identified, and these two groups are those of vowel and consonant. If we look at the vowel–consonant distinction in this way, we must say that the most important difference between vowel and consonant is not the way that they are made, but their different distributions. It is important to remember that the distribution of vowels and consonants is different for each language.

We begin the study of English sounds in this course by looking at vowels, and it is necessary to say something about vowels in general before turning to the vowels of English. We need to know in what ways vowels differ from each other. The first matter to consider is the shape and position of the tongue. It is usual to simplify the very complex possibilities by describing just two things: firstly, the vertical distance between the upper surface of the tongue and the palate and, secondly, the part of the tongue, between front and back, which is raised highest. Let us look at some examples:

i) Make a vowel like the iː in the English word 'see' and look in a mirror; if you tilt your head back slightly you will be able to see that the tongue is held up close to the roof of the mouth. Now make an æ vowel (as in the word 'cat') and notice how the distance between the surface of the tongue and the roof of the mouth is now much greater. The difference between iː and æ is a difference of tongue height, and we would describe iː as a relatively **close** vowel and æ as a relatively **open** vowel. Tongue height can be changed by moving the tongue up or down, or moving the lower jaw up or down. Usually we use some combination of the two sorts of movement, but when drawing side-of-the-head diagrams such as Fig. 1 and Fig. 2 it is usually found simpler to illustrate tongue shapes for vowels as if tongue height were altered by tongue movement alone, without any accompanying jaw movement. So we would illustrate the tongue height difference between iː and æ as in Fig. 3.

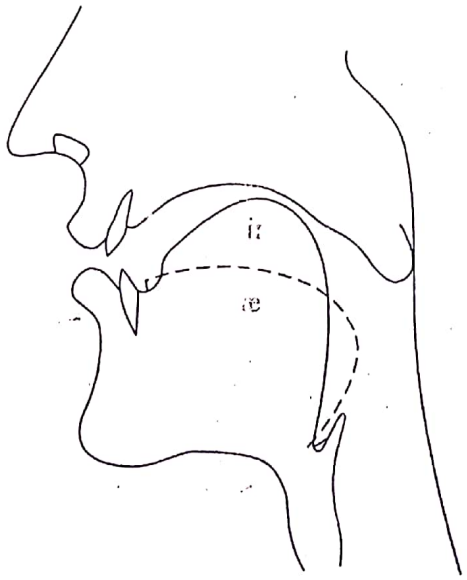

Fig. 3 Tongue positions for iː and æ

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ii) In making the two vowels described above, it is the front part of the tongue that is raised. We could therefore describe iː and æ as comparatively **front** vowels. By changing the shape of the tongue we can produce vowels in which a different part of the tongue is the highest point. A vowel in which the back of the tongue is the highest point is called a **back** vowel. If you make the vowel in the word 'calm', which we write phonetically as ɑː, you can see that the back of the tongue is raised. Compare this with æ in front of a mirror; æ is a front vowel and ɑː is a back vowel. The vowel in 'too' (uː) is also a comparatively back vowel, but compared with ɑː it is close.

So now we have seen how four vowels differ from each other; we can show this in a simple diagram.

	Front	Back
Close	iː	uː
Open	æ	ɑː

However, this diagram is rather inaccurate. Phoneticians need a very accurate way of classifying vowels, and have developed a set of vowels which are arranged in a close–open, front–back diagram similar to the one above but which are not the vowels of any particular language. These **cardinal vowels** are a standard reference system, and people being trained in phonetics at an advanced level have to learn to make them accurately and recognise them correctly. If you learn the cardinal vowels, you are not learning to make English sounds, but you are learning about the range of vowels that the human vocal apparatus can make, and also learning a useful way of describing, classifying and comparing vowels. They are recorded on CD 2, Track 12.

It has become traditional to locate cardinal vowels on a four-sided figure (a quadrilateral of the shape seen in Fig. 4 – the design used here is the one recommended by the **International Phonetic Association**). The exact shape is not really important – a square would do quite well – but we will use the traditional shape. The vowels in Fig. 4 are the so-called **primary** cardinal vowels; these are the vowels that are most familiar to the speakers of most European languages, and there are other cardinal vowels (**secondary** cardinal vowels) that sound less familiar. In this course cardinal vowels are printed within square brackets [ ] to distinguish them clearly from English vowel sounds.

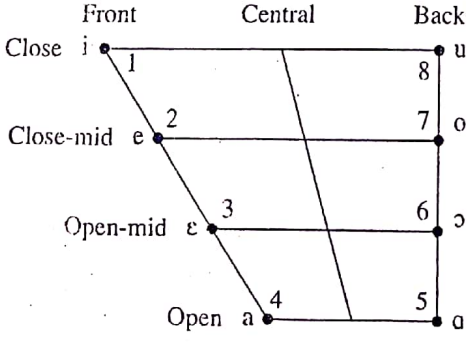

Fig. 4 Primary cardinal vowels

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Cardinal vowel no. 1 has the symbol [i], and is defined as the vowel which is as close and as front as it is possible to make a vowel without obstructing the flow of air enough to produce friction noise; friction noise is the hissing sound that one hears in consonants like s or f. Cardinal vowel no. 5 has the symbol [ɑ] and is defined as the most open and back vowel that it is possible to make. Cardinal vowel no. 8 [u] is fully close and back and no. 4 [a] is fully open and front. After establishing these extreme points, it is possible to put in intermediate points (vowels no. 2, 3, 6 and 7). Many students when they hear these vowels find that they sound strange and exaggerated; you must remember that they are *extremes* of vowel quality. It is useful to think of the cardinal vowel framework like a map of an area or country that you are interested in. If the map is to be useful to you it must cover all the area; but if it covers the whole area of interest it must inevitably go a little way beyond that and include some places that you might never want to go to.

When you are familiar with these extreme vowels, you have (as mentioned above) learned a way of describing, classifying and comparing vowels. For example, we can say that the English vowel æ (the vowel in 'cat') is not as open as cardinal vowel no. 4 [a]. We have now looked at how we can classify vowels according to their tongue height and their frontness or backness. There is another important variable of vowel quality, and that is lip-position. Although the lips can have many different shapes and positions, we will at this stage consider only three possibilities. These are:

i) **Rounded**, where the corners of the lips are brought towards each other and the lips pushed forwards. This is most clearly seen in cardinal vowel no. 8 [u].
ii) **Spread**, with the corners of the lips moved away from each other, as for a smile. This is most clearly seen in cardinal vowel no. 1 [i].
iii) **Neutral**, where the lips are not noticeably rounded or spread. The noise most English people make when they are hesitating (written 'er') has neutral lip position.

Now, using the principles that have just been explained, we will examine some of the English vowels.

2.3 English short vowels

AU2, Exs 1–5

English has a large number of vowel sounds; the first ones to be examined are short vowels. The symbols for these short vowels are: ɪ, e, æ, ʌ, ɒ, ʊ. Short vowels are only *relatively* short; as we shall see later, vowels can have quite different lengths in different contexts.

Each vowel is described in relation to the cardinal vowels.

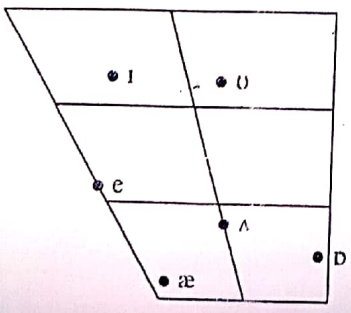

Fig. 5 English short vowels

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CHAPTER 5

The Classification and Description of Speech Sounds II: Vowels

We said in the last chapter that during the articulation of vowel sounds, the lung-air escapes through the mouth without any friction. This is because during the articulation of vowels there is no obstruction in the mouth, (as there *is* during the articulation of plosives, nasals and affricates) nor is there any narrowing in the mouth to allow friction (as there *is* during the articulation of fricative consonants). Vowels are therefore articulated with a stricture of *open approximation*. That is to say, the active articulator is raised in the direction of the passive articulator in such a way that there is sufficient gap between them to allow the air to escape freely and continuously, *without any friction*. (Vowels are thus articulated exactly like the consonants we termed **approximants** in the previous chapter. We shall discuss the difference between vowels and the consonants called *approximants* in a subsequent chapter.)

The active articulator during the articulation of all vowels (except in the case of retroflex vowels or r-coloured vowels) is the front of the tongue or the back of the tongue or a part of the tongue that is between the front and back of the tongue which we call the *centre* of the tongue. The passive articulator during the articulation of all vowels is either the hard palate (if the front of the tongue is the active articulator) or the soft palate (if the back of the tongue is the active articulator) or that part of the roof of the mouth between the hard palate and the soft palate (if the 'centre' of the tongue is the active articulator).

Let us illustrate this point. Say the English word *bead* slowly. You will realize that during the articulation of the sound represented by the letters *ea* the air escapes through the mouth freely and continuously without any friction. Now say the English words *bad*, *bath* and *booed*. You will realize that during the articulation of the sounds represented by the letters *a*, *a* and *oo* in these words, the air once again escapes freely and continuously through the mouth, without

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any friction. So the sounds represented by the letters *ea* in *bead*, *a* in *bad*, *a* in *bath* and *oo* in *booed* are all vowels. Now say these words again slowly, prolonging the vowel sounds. You will realise that though all the four sounds referred to above are vowels, each one sounds different from every other. This is because during the articulation of these four sounds the tongue takes up different positions in the mouth and consequently the cavity of the mouth through which the lung-air passes is different each time. Each of these vowels (and in fact every vowel) has a distinct *quality* and this is because of the different positions taken up by the tongue during different vowel articulations. It therefore becomes necessary to classify vowels and decribe them.

The tongue can assume a large number of positions during the articulation of vowel sounds, but the upper surface of the main body of the tongue is usually *convex*. This is because some part of the tongue—the front, back or centre—is raised in the direction of the roof of the mouth.

Let us examine the diagram that is given below.

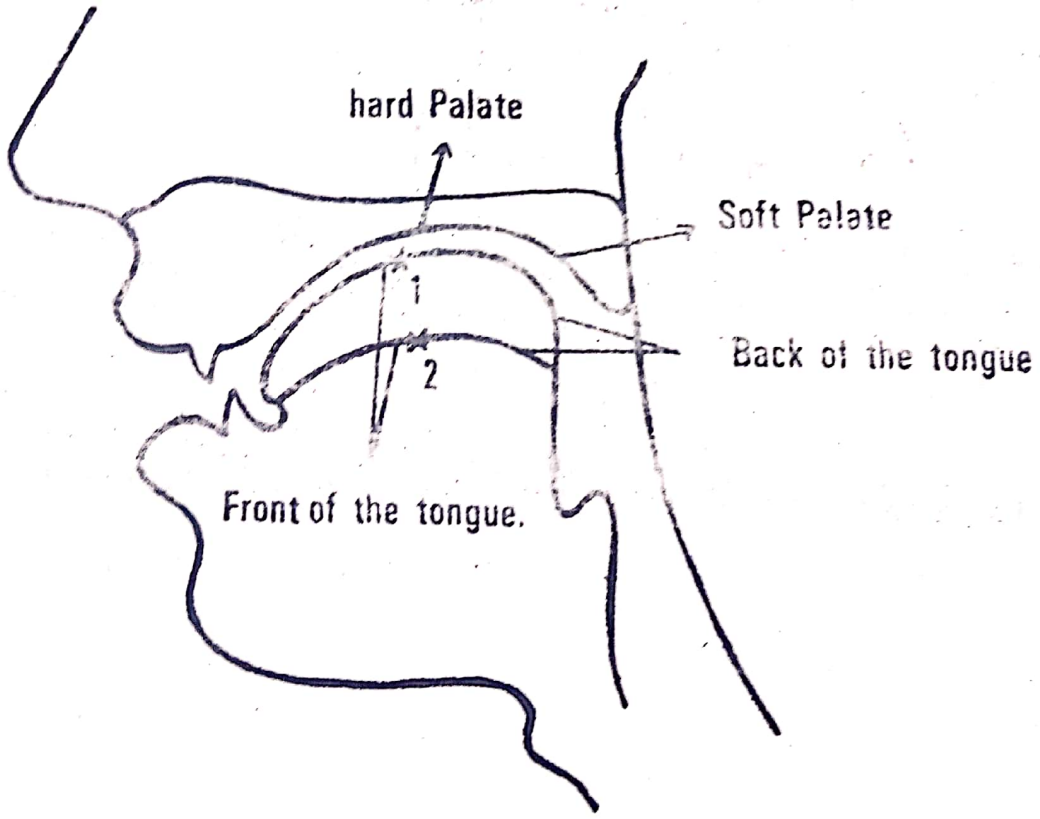

Fig. 30

Two different tongue-positions are marked (with the numerals 1 and 2) in Figure 30 given above. We have to imagine that in both the cases the subject is articulating a vowel sound, i.e., we have to imagine that in both the cases the space between the highest point

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of the tongue (marked with a cross) and the roof of the mouth is wide enough for the air to escape without any friction. In both the cases the front of the tongue is raised in the direction of the hard palate. So we can say that the speaker in Figure 30 is producing two different **front vowels.** *Front vowels are those during the articulation of which the* **front** *of the tongue is raised in the direction of the hard palate.* It should be remembered, however, that the space between the front of the tongue and the hard palate should be sufficient for the air to escape without any friction. The vowels in the English words *bee, bid, bed* and *bad* are some examples of front vowels.

Now let us look at Figure 31 given below:

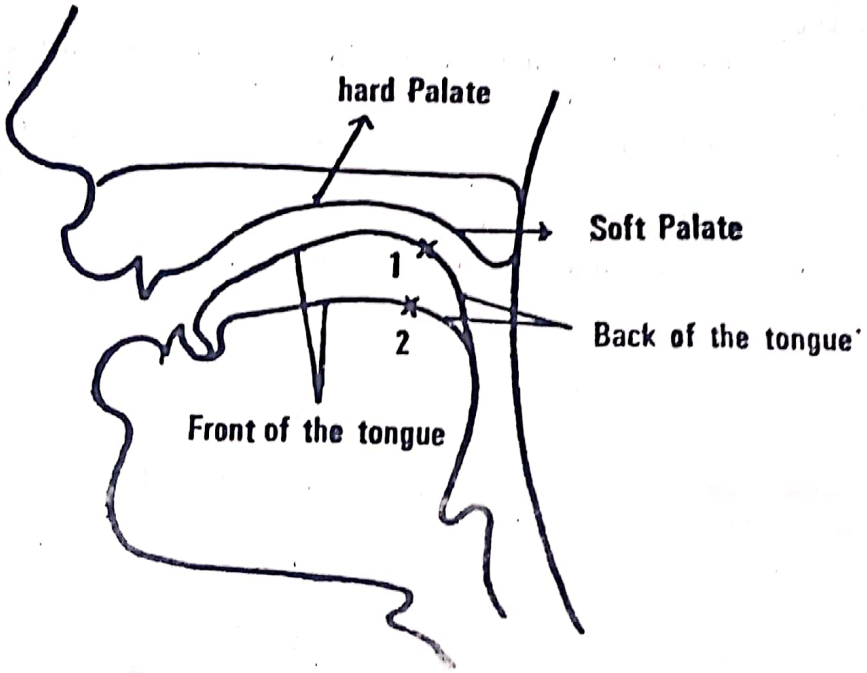

Fig. 31

As in Figure 30, two different tongue-positions are marked (again with the numerals 1 and 2). We have to assume once again that the subject is articulating vowel sounds. In other words, we have to imagine that the space between the highest point of the tongue (marked with a cross) and the roof of the mouth is sufficient for the air to escape without any friction. In this figure, in the case of either of the two tongue-positions marked, the back of the tongue is raised in the direction of the soft palate. We can therefore say that the speaker in Figure 31 is producing two different **back vowels.** *Back vowles are those during the articulation of which the* **back** *of the tongue is raised in the direction of the soft palate in such a way that*

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there is a sufficiently wide gap between them for the air to escape without friction. The vowels in the English words *cart, cot, caught, push* and *pool* are some examples of back vowels.

Vowels may be produced by raising the centre of the tongue in the direction of that part of the roof of the mouth where the hard palate and the soft palate meet. These vowels are called central vowels. The vowels in the English words *cup* and *heard* are some examples of central vowels.

The above discussion leads us to the conclusion that vowels can be classified into front, back and central, taking into consideration the part of the tongue that is highest in the mouth during their articulation.

If we look at Figures 30 and 31 again, we will see that each of the figures illustrates two vowel articulations. Figure 30 illustrates the articulation of two front vowels (since during the articulation of either of these, the front of the tongue is raised in the direction of the hard palate) whereas Figure 31 illustrates the articulation of two back vowels (since during the articulation of either of these, the back of the tongue is raised in the direction of the soft palate). Since each of these two figures illustrates the articulation of two vowels with the same part of the tongue raised (there are more possibilities than the two represented in these figures), this sort of classification of vowels according to the part of the tongue raised is not sufficient. Vowels will have to be classified still further. We must be able to distinguish between the vowels whose tongue-positions are marked 1 and 2 in Figure 30 using some other criterion since both are front vowels. Similarly we have to find other means of distinguishing between the vowels whose tongue-positions are marked 1 and 2 in Fig. 31 since both of them are back vowels.

Let us look carefully at Figures 30 and 31 again. In Figure 30, during the articulation of the vowel whose tongue-position is marked 1, the front of the tongue is quite close to the roof of the mouth (the hard palate, to be precise). We said earlier on that we have to assume that the space between the point marked 1 and the hard palate is sufficient for the air to escape without any friction. Let us assume one more thing now, viz., that the point marked 1 is the highest to which the front of the tongue can be raised in order to produce a vowel. That is to say, if we raise the front of the tongue higher than the point marked 1, the space between the two articulators will become so narrow as to produce friction. In that case, the sound

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produced will not be a vowel. (Remember that the front of the tongue can be raised higher than the point marked 1 in Figure 30. In fact the front of the tongue can be raised till it touches the hard palate. But what we are assuming is that if the gap between the front of the tongue and the hard palate is decreased any further than indicated in Figure 30, the air will escape through the gap with friction and therefore the sound produced will not be a vowel sound.) Similarly in Figure 31, the back of the tongue is quite close to the roof of the mouth. Once again let us assume that the point marked 1 is the highest the back of the tongue can be raised so as to produce a vowel sound, i.e., if the back of the tongue is raised any further, the space between the back of the tongue and the soft palate will become so narrow that the lung-air will escape through this gap with friction. The vowels whose tongue-positions are marked 1 in Figures 30 and 31 are close vowels, because during their articulation, the tongue (the front of the tongue in the case of Figure 30 and the back of the tongue in the case of Figure 31) is very close to the roof of the mouth, consistent with the sound produced being a vowel. The vowel whose tongue-position is marked 1 in Figure 30 is a **front close vowel** (combining the two criteria, viz., the part of the tongue that is raised and the height to which it is raised) and the vowel whose tongue-position is marked 1 in Figure 31 is a **back close vowel.**

Now let us look at the vowels whose tongue-positions are marked 2 in Figures 30 and 31. In either case, the tongue (front in Figure 30 and back in Figure 31) is far away from the roof of the mouth. Let us assume that the front of the tongue cannot be lowered any further than the point marked 2 in figure 30 and that if the back of the tongue is lowered any further than the point marked 2 in Figure 31, friction will be heard and thus the sound produced will not be a vowel. The vowels marked 2 in the two figures can be called open vowels because during their articulation, the tongue is far away from the roof of the mouth. So we can have front close vowels, central close vowels, back close vowels, front open vowels, central open vowels and back open vowels.

Let us now look at Figure 32 given below. Four different tongue-positions are marked in the diagram. We are familiar with the tongue-positions marked 1 and 4. Number 1 is a **front close vowel** (the front of the tongue is raised very close to the hard palate) and 4 is a **front open vowel** (again the front of the tongue is raised, but it is far away from the hard palate). These are two

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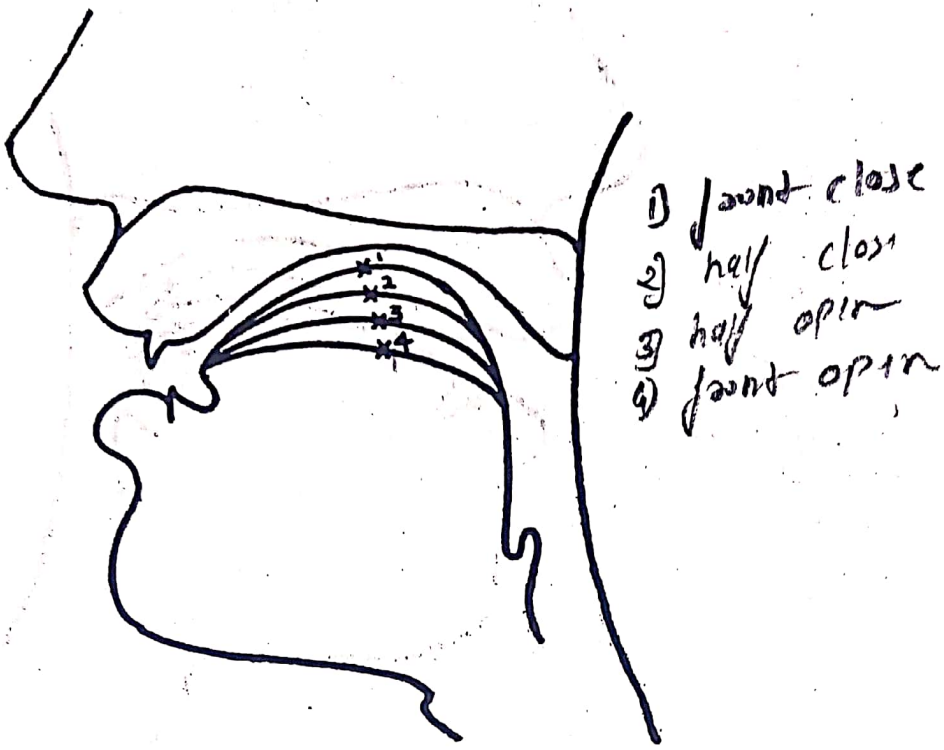

Fig. 32

extreme points. (If we raise the front of the tongue higher than the point marked 1 the air will escape with friction and the sound produced will thus not be a vowel; and it is not possible to lower the front of the tongue beyond the point marked 4.) It is possible that the front of the tongue lies anywhere between these two extreme points during the articulation of the vowels of any given language. For a theoretical description, the distance between these extreme positions (i.e., the close and open positions) is divided into three equal parts as indicated in Figure 32. The position marked 2 in Figure 32 is called half-close (since the tongue is between the close and open positions, but closer to the close position than to the open position). The position marked 3 in Figure 32 is called **half-open** (since the tongue is between the close and open positions, but closer to open than to close this time).

Now let us look at Figure 33 given below. We are once again familiar with the vowels marked 8 and 5. Number 8 is a **back close** vowel (the *back* of the tongue is raised *close* to the soft palate during its articulation) and number 5 is a **back open vowel** (the back of the tongue is raised but it is far away from the soft palate

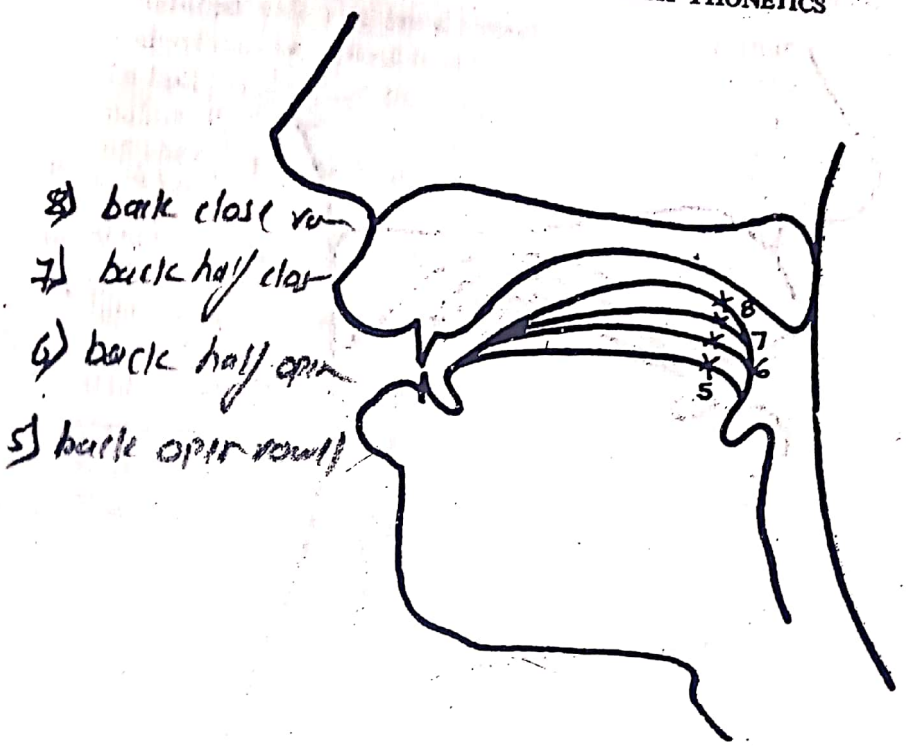

Fig. 33

during the articulation of this vowel). The back of the tongue can be anywhere between these two extreme points during the articulation of the vowels of any given language. The position marked 7 in Figure 33 is called **back half-close** and the position marked 6 is called **back half-open.**

Similarly, we can have central half-close and central half-open vowels in which cases the 'centre' of the tongue will be raised to a point between the close and open positions.

So far we have considered two criteria for the description and classification of vowels. They are:

1. the part of the tongue raised;
2. the height to which it is raised.

and these give us 12 theoretical possibilities—a threefold classification according to the former and a fourfold classification according to the latter.

A very important point to be remembered is this. What we have described so far are *theoretical possibilities.* During the articulation

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of a vowel in any language, the tongue (front, back or centre) may actually lie anywhere between the close and open positions. The tongue may be between close and half-close, but closer to close; it may be exactly midway between close and half-close; it may be between close and half-close, but closer to half-close; it may be between half-close and half-open, closer to half-close—it may be anywhere between the two extreme points. What we have called close, half-close, half-open and open are *convenient reference points* with reference to which any vowel in any language may be described.

Yet another criterion we take into account for the classification of vowels is the *position of the lips*. According to this, vowels are divided into two categories—**unrounded** and **rounded.** Unrounded vowels are those during the articulation of which the lips are spread or neutral. Rounded vowels are those during the articulation of which the lips are rounded. The vowels in the English words *seat, sit, set* and *sat* are *unrounded vowels* and those in *cot, caught, put* and *fool* are *rounded vowels.* There are various degrees of lip-spreading and lip-rounding, but for the sake of convenience, vowels are classified as unrounded and rounded.

To summarise, we describe a vowel taking into account the following criteria:

(a) the part of the tongue raised during its articulation (front, central, back);
(b) the height to which it is raised (close, half-close, half-open and open);
(c) the position of the lips (unrounded and rounded).

These give us $3 \times 4 \times 2 = 24$ theoretical possibilities.

We can describe a vowel using a short three-term descriptive label, the three terms indicating the three criteria we have considered so far in our description of vowels. Thus a *back half-open rounded* vowel would mean the following:

(a) the back of the tongue is raised in the direction of the soft palate during its articulation;
(b) it is raised to a point between the close and open positions, but closer to open than to close;
(c) the lips are rounded during its articulation.

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Fig. 34

We are now familiar with tongue-positions marked 1 to 8 in the figure. The area joining the eight points is called the **vowel area.** We can articulate vowel sounds if the tongue falls somewhere within this vowel area. The vowels articulated with these eight tongue-positions are called **Cardinal Vowels.** Cardinal vowels 1 to 5 (i.e., front close, front half-close, front half-open, front open and back open) are articulated with various degrees of lip-spreading and Cardinal vowels 6 to 8 (i.e., back half-open, back half-close and back close) are articulated with various degrees of lip-rounding. The Cardinal vowels are described below with a three-term label for each and the phonetic symbol allotted to it by the alphabet of the International Phonetic Association:

Cardinal vowel no. 1 front close unrounded—symbol [i]
Cardinal vowel no. 2 front half-close unrounded—symbol [e]
Cardinal vowel no. 3 front half-open unrounded—symbol [ɛ]
Cardinal vowel no. 4 front open unrounded—symbol [a]
Cardinal vowel no. 5 back open unrounded—symbol [ɑ]
Cardinal vowel no. 6 back half-open rounded—symbol [ɔ]

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Cardinal vowel no 7 back half-close rounded—symbol [o]
Cardinal vowel no 8 back close rounded—symbol [u]

In fact it is possible to produce vowels with exactly the same tongue-positions as for the eight cardinal vowels described above, but with the lip-position reversed. In other words, it is possible to have front close, half-close, half-open, open and back open **rounded** vowels and back half-open, half-close and close **unrounded** vowels. These eight are called **secondary cardinal** vowels. Since English does not have any front rounded vowels nor back unrounded vowels in the close, half-close and half-open positions, we shall not discuss the secondary cardinal vowels any further.

The vowel diagram: In most books dealing with phonetics you will find a quadrilateral to indicate the tongue-positions during the articulation of vowels. The quadrilateral looks like the one given below and it is obtained by joining the points numbered 1 to 8 in Figure 34.

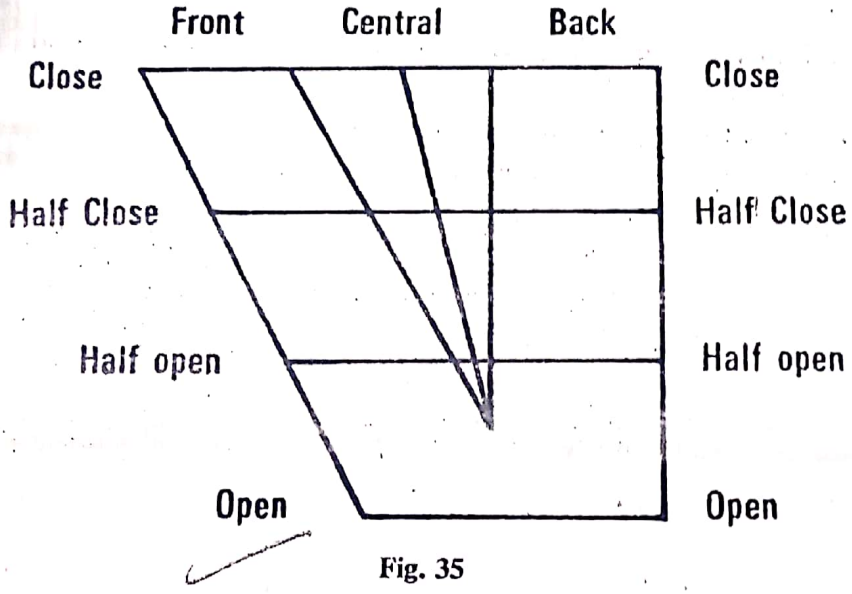

Fig. 35

It is very important to remember what we have already stated elsewhere, in this chapter, viz., that the cardinal vowels do not exist in any particular language. These are reference points which may be used as a yardstick to describe the actual vowels present in a language. We can say, for example, that the vowel in the English word *see* is very near cardinal vowel 1, that the vowel in the English word *shoe* is very near cardinal vowel 8 and so on.

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CHAPTER 14

Word-Accent

The word is a linguistic entity composed of one or more phonemes. The words *I*, *oh*, consist of one phoneme each—the diphthong /ai/ and /əu/ respectively. The words *bee* and *saw* have two phonemes each—/b/ and /i:/ and /s/ and /ɔ:/ respectively. The words *cat* and *big* have three phonemes each—/k/, /æ/ and /t/ and b/, /i/ and /g/ respectively. The words *crack* and *trein* have four phonemes each—/k/, /r/, /æ/ and /k/ and /t/, /r/, /ei/ and /n/ respectively. The words *blast* and *friend* have five phonemes each—/b/, /l/, /ɑ:/ /s/ and /t/ and /f/, /r/ /e/, /n/ and /d/ Though the words cited above have different number of phonemes each, all of them have one syllable each—all of them are monosyllabic words.

There are a number of words in every language which have more than one syllable. The English words *tailor* and *teacher* have two syllables each. The second syllable in these words have the vowel /ə/. We can therefore say (since /ə/ in English does not occur in an accented syllable—more is said about accented syllables in this chapter) that these words have a strong syllable and a *weak* syllable, *in that order*. If we symbolise a strong syllable with a heavy dot /●/ and a weak syllable with a light dot /./, we can symbolise words such as the ones cited above as /●./. Words like *again* and *perceive* have a weak syllable and a strong syllable, arranged in that order. Words such as these can be symbolised /.●/. In an English word of more than one syllable, one of the syllables is pronounced with *greater prominence* than the other(s). The syllable that is pronounced more prominently than the other(s) in the same word is said to be *accented* or *to receive the accent*.

The greater prominence of a syllable may be due to *stress* or greater breath force, but often the length of the vowel in a syllable, stress and pitch change (pitch will be discussed in a later chapter) work together to render a syllable more prominent than its neighbouring syllables.

In several English words consisting of several syllables each, more

than one syllable may be prominent. For example, in the word *Examination* /ig-zæ-mi-nei-ʃn/ which has five syllables, the syllable that has the maximum prominence is the fourth syllable /nei/, the syllable that has the next degree of prominence is the second syllable /zæ/; the other syllables /ig/, /mi/ and /ʃn/ will be pronounced with less prominence than the two syllables listed above. Of these three syllables /ig/ will have the maximum prominence (but less than /nei/ and /zæ/) and /ʃn/ the minimum prominence and /mi/ will have prominence in between these two. So this word has two prominent or strong syllables—/nei/ and /zæ/. Of these two, a listener will hear /nei/ as being more prominent because, apart from this syllable being articulated with greater breath force, *there will be a pitch movement on this syllable*. In a polysyllabic word the syllable on which a pitch movement takes place is said to receive *primary accent* or *tonic accent*. Any other prominent syllable in the same word is said to receive *secondary* accent.

Different dictionaries mark word-accent in different ways. The method used in most modern books on linguistics and phonetics (including Daniel Jones's *An English Pronouncing Dictionary* and the *Oxford Advanced Learner's Dictionary*) is the following. The same method has been adopted in this book.

Primary accent is marked with a vertical bar *above and in front of the* syllable to which it refers. Secondary accent is marked with a vertical bar *below* and *in front of* the syllable to which it refers. A few English words are given below in phonemic transcription with both primary and secondary accents marked:

categorical	/ˌkæti'gɒrikl/
comprehensible	/ˌkɒmpri'hensəbl/
conventionality	/kənˌvenʃə'næliti/
deposition	/ˌdepə'ziʃn/
examination	/igˌzæmi'neiʃn/
international	/ˌintə'næʃənl/
interpolation	/inˌtɜ:pə'leiʃn/
notification	/ˌnəutifi'keiʃn/

What we have said so far is very simple. In an English word of more than one syllable, one of the syllables stands out from the rest and this syllable is said to receive the accent. There are several polysyllabic words in which more than one syllable in each word

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may be prominent—one of the syllables is said to receive the primary or the tonic accent; the other syllable that is prominent (i.e., a syllable that sounds prominent in spite of there being no pitch movement on it) is said to receive secondary accent.

What is more difficult for non-native speakers of English is to place the accent on the correct syllable(s) of a word. In other words, which syllable(s) of a polysyllabic word is/are to be pronounced with greater prominence? This is a difficult question to answer because in English words the accent is *both free and fixed.* It is free in the sense that it is not associated with a particular syllable of a word. In other words, there are a number of disyllabic words like *'teacher 'wisdom, 'cancel* and *'doctor* which receive the accent on the first syllable; there are a number of disyllabic words like *be'come, can'teen, re'mark* and *se'rene* which receive the accent on the second syllable. There are trisyllabic words like *'advertise, 'anything 'beautiful* and *'customer* which receive the accent on the *first syllable*, like *a'greement, ar'tistic* and *de'velop* which receive the accent on the *second syllables*; like *ciga'rette, disap'point* and *under'stand* which receive the accent on the *third syllable*. The accent in English words is fixed in the sense that the main accent always falls on a particular syllable. (There are a few exceptions and these are determined by the rhythmic pattern of the sentence in which the words occur. Rhythm will be discussed in the next chapter.) Since the accent in English words is not associated with a particular syllable, a non-native speaker of English will have to learn this very important aspect of spoken English with great care. He has to learn which syllable of a polysyllabic word is accented by referring to a good dictionary.

There is a second difficulty about English word-accent and this is owing to a shift in the accented syllable. In derivatives a shift in the accent is quite common. There are, no doubt, a number of words like *a'gree* and *a'greement* in which the derived word takes the accent on the same syllable on which the word from which it is derived takes the accent. But there are a number of derivatives in which there is a change in the accented syllable. A few examples are given below:

a'cademy (primary accent on the *second* syllable)
aca'demic (Primary accent on the *third* syllable)
acade'mician (primary accent on the *fourth* syllable)

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'advertise	(primary accent on the *first* syllable)
ad'vertisement	(primary accent on the *second* syllable)
ex'amine	(primary accent on the *second* syllable)
exami'nee	(primary accent on the *third* syllable)
exami'nation	(primary accent on the *fourth* syllable)
in'ferior	(primary accent on the *second* syllable)
inferi'ority	(primary accent on the *fourth* syllable)
'photograph	(primary accent on the *first* syllable)
pho'tographer	(primary accent on the *second* syllable)
photo'graphic	(primary accent on the *third* syllable)
'politics	(primary accent on the *first* syllable)
po'litical	(primary accent on the *second* syllable)
poli'tician	(primary accent on the *third* syllable)
res'ponsible	(primary accent on the *second* syllable)
responsi'bility	(primary accent on the *fourth* syllable)

There is yet another aspect of English word-accent. There are a number of disyllabic words in English in which word-accent depends upon whether the words are used as nouns/adjectives or as verbs. If these words are used as nouns or adjectives, the accent is on the *first* syllable and if these are used as verbs, the accent is on the *second* syllable. A few of these are listed below:

Word	*Noun or Adjective*	*Verb*
absent	/'æbsənt/	/əb'sent/
concert	/'kɔnsət/	/kən'sə:t/
conduct	/'kɔn,dʌkt/	/kən'dʌkt/
contact	/'kɔn,tækt/	/kən'tækt/
contract	/'kɔn,trækt/	/kən'trækt/
contrast	/'kɔn,trɑ:st/	/kən'trɑ:st/
convict	/'kɔnvikt/	/kən'vikt/
desert	/'dezət/	/di'zə:t/
digest	/'daidʒest/	/di'dʒest/or /dai'dʒest/
export	/'ekspɔ:t/	/iks'pɔ:t/
import	/'impɔ:t/	/im'pɔ:t/
object	/'ɔbdʒikt/	/əb'dʒekt/
perfect	/'pə:fikt/	/pə'fekt/
permit	/'pə:mit/	/pə'mit/
present	/'preznt/	/pri'zent/
produce	/'prɔ,dju:s/	/prə'dju:s/

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It is to be remembered, however, that not all disyllabic words which can be used as nouns and verbs undergo such a shift in the accented syllable. There are words like *'limit 'order, re'mark, 'visit*, etc. which are accented on the same syllable whether they are used as nouns or as verbs.

Accent in Compound Words: By *compound word* is meant a word composed of two separable words. In spelling them, there may or may not be a hyphen between the two elements forming the compound.

i) In most compound words in English the primary accent falls on one of the two elements. The most common type in English is the *first* of the two elements receiving the primary accent. A few examples are given below:

'air-raid	'lifeboat
'blackbird	'mailbag
'bookshelf	'offprint
'cardboard	'pickpocket
'coalman	'postman
'crossword	'rain-coat
'footprint	'school-bus
'hairbrush	'tea-party

ii) There are, however, a few compound words with *-ever* and *-self* as the second elements in which the second element receives the primary accent. A few examples as given below:

her'self	how'ever
him'self	what'ever
my'self	when'ever
them'selves	who'ever

iii) There are other compound words in which both the elements are accented, but the primary accent falls on the second element. For example:

ˌafter-'noon	ˌhome-'made
ˌbad-'tempered	ˌpost-'graduate
ˌcountry-'house	ˌvice-'chancellor
ˌgood-'looking	

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The foregoing account may give one the impression that word-accent in English is something that is irregular and unpredictable. There are, however, a few useful rules for word-accentual patterns in English. These are given below with examples.

Rule 1: Words with weak prefixes always take the accent on the root. For example,

a'board	/ə'bɔ:d/
a'broad	/ə'brɔ:d/
a'head	/ə'hed/
a'lone	/ə'ləun/
be'cause	/bi'kɔz/
be'come	/bi'kʌm/
be'low	/bi'ləu/
be'neath	/bi'ni:θ/

Rule 2: The *inflexional* suffixes *-ed, -es* and *-ing* do not affect the accent. For example,

-ed	*recom'mend*	*recom'mended*
	re'late	*re'lated*
	so'licit	*so'licited*
	sub'mit	*sub'mitted*
-es	*com'pose*	*com'poses*
	dis'ease	*dis'eases*
	'focus	*'focusses*
	suc'cess	*suc'cesses*
-ing	*ad'vance*	*ad'vancing*
	com'mit	*com'mitting*
	'happen	*'happening*
	'reason	*'reasoning*

Rule 3: The *derivational* suffixes *-age, -ance, -en, -er, -ess, -ful, -hood, -ice, -ish, -ive, -less, -ly, -ment, -ness, -or, -ship, -ter, -ure* and *-zen* do not normally affect the accent. For example,

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-age	*'carry*	*'carriage*
	'cover	*'coverage*
	'hermit	*'hermitage*
	'marry	*'marriage*
-ance	*an'noy*	*an'noyance*
	ap'pear	*ap'pearance*
	at'tend	*at'tendance*
	per'form	*per'formance*
-en	*'bright*	*'brighten*
	'light	*'lighten*
-er	*at'tend*	*at'tender*
	be'gin	*be'ginner*
	'common	*'commoner*
	per'form	*per'former*
-ess	*'actor*	*'actress*
	'author	*'authoress*
	con'ductor	*con'ductress*
	'tiger	*'tigress*

(But *prince—prin'cess* except when used attributively when it is *'princess*)

-ful	*'beauty*	*'beautiful*
	'colour	*'colourful*
	'duty	*'dutiful*
	re'venge	*re'vengeful*
-hood	*'brother*	*'brotherhood*
	'priest	*'priestgood*
	'sister	*'sisterhood*
	'widow	*'widowhood*
-ice	*'coward*	*'cowardice*
ish	*'fever*	*'feverish*
	'white	*'whitish*
	'woman	*'womanish*
	'yellow	*'yellowish*

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-ive	*a'buse*	*a'busive*
	at'tend	*at'tentive*
	con'clude	*con'clusive*
	per'mit(v)	*per'missive*
-less	*'bottom*	*'bottomless*
	'colour	*'colourless*
	'manner	*'mannerless*
	'odour	*'odourless*
-ly	*'certain*	*'certainly*
	'purpose	*'purposely*
	'tactless	*'tactlessly*
	'wanton	*'wantonly*
-ment	*a'chieve*	*a'chievement*
	con'tent	*con'tentment*
	in'volve	*in'volvement*
	'manage	*'management*
-ness	*'bitter*	*'bitterness*
	'common	*'commonness*
	'lovely	*'loveliness*
	'useless	*'uselessness*
-or	*col'lect*	*col'lector*
	con'duct (v)	*con'ductor*
	op'press	*op'pressor*
	pos'sess	*pos'sessor*
-ship	*'author*	*'authorship*
	'scholar	*'scholarship*
	'student	*'studentship*
	'warden	*'wardenship*
-ter	*'laugh*	*'laughter*
-zen	*'city*	*'citizen*

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Rule 4: Words ending in *-ion* take the primary accent on the *penultimate* syllable. Here are a few examples:

ˌadmi'ration	ˌfalsifi'cation
ˌappli'cation	'motion
ˌcombi'nation	'nation
ˌdeco'ration	'notion
deˌtermi'nation	ˌprepa'ration
exˌami'nation	'station

Rule 5: Words ending in *-ic, -ical, -ically, -ious, -ial* and *-ially* take the primary accent on the *syllable preceding the suffix*. For example,

-ic
- aˌpolo'getic
- pa'thetic
- ˌsympa'thetic
- ter'rific

-ical
- bio'logical
- e'lectrical
- 'optical
- ˌpsychol'ogical

-ically
- 'chemically
- ˌeco'nomically
- ˌpsycho'logically
- sta'tistically

-ious
- a'trocious
- ˌcere'monious
- fal'lacious
- no'torious

-ial
- com'mercial
- ˌconfi'dential
- ˌdicta'torial
- me'morial

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-ially	*ˌcate'gorically*
	com'mercially
	ˌconfi'dentially
	dra'matically

Rule 6: Words ending in *-ity* take the accent on the *ante-penultimate* syllable, or the third syllable from the end. For example,

a'bility	*ˌgene'rosity*
ca'pacity	*ˌmagna'nimity*
elec'tricity	*o'pacity*
e'normity	*ˌoppor'tunity*
fu'tility	*ratio'nality*

independence

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